28-38 - Booth No. 48: HOW BIG DATA WORLD IS AFFECTING GEOSCIENCE EDUCATION: A STUDY IN HIGH-RANKED US-COLLEGES

Saturday, 18 March 2023

8:00 AM - 12:00 PM

Grand Ballroom A–D (Hyatt Regency Reston)

Booth No. 48

Abstract

The shift of our world from a service-based, industrial society to a knowledge-based, big-data world has proven to touch the lives of people worldwide. Geoinformatics is an emerging field that facilitates this transition within the earth-science community. Geoinformatics refers to the science and technology used to develop information science infrastructure to address problems and interpretations of large-scale geological phenomena. The benefits of geoinformatics are constantly growing as scientific and technological advances in Artificial Intelligence and large data sets queries continue to develop exponentially, yet the education surrounding geoinformatics fails to retain the same pace. A manual search for geoinformatics courses in the course catalogs of U.S. colleges within the best geology departments ranked by U.S. News revealed that these top programs have little to no courses involving geoinformatics. However, in ranking the schools in order of geoinformatics courses offered within their geology courses, it was found that the Western colleges provided more geoinformatics courses than the Eastern colleges with few exceptions. This phenomenon can be understood by looking at the industry composition of the West in comparison to the East. For example, Silicon Valley, a region that serves as a global center for advanced technology and innovation, along with several other big data headquarters, such as Google, Facebook, Linkdn, are located in the Western U.S. Although education surrounding geoinformatics is sparse, the literature review finds that knowledge in the form of geoinformatics is of vital importance and provides benefits in several geo-applications and disciplines. Thus, the U.S. education system should shift focus towards geoinformatics gaining prominence within the school earth-science education in the U.S. The implications of this research prompt an urgent need for a detailed scholarly analysis of research in this area and curation of a standardized curriculum for geoinformatics within the U.S. college system. The permeation of geoinformatics through these methods would properly equip our future geologists with the knowledge required for the big data world needs of 2050.

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